

NUMERICAL CONSTRUCTION OF THE EVEN ICOSAHEDRAL MAASS FORM OF CONDUCTOR 1951

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ABSTRACT. Let ρ be the even icosahedral Artin representation of prime conductor 1951 found by Doud and Moore, the smallest prime conductor for which such a representation exists. Artin’s holomorphy conjecture for ρ is equivalent to the existence of a Maass newform of level 1951, nebentypus the quintic character χ_{780} , and Laplace eigenvalue $1/4$, whose Hecke eigenvalues are the traces $\text{tr } \rho(\text{Frob}_p)$. We report a numerical construction of that form by Hejhal’s algorithm, together with a battery of independent consistency checks. The Fourier coefficients reproduce the Frobenius classes of the Doud–Moore quintic at all 549 unramified primes below 4000; after untwisting by χ_{780}^3 they are real and lie in $\mathbb{Z}[\varphi]$ at all 3998 indices below 4000; they satisfy Hecke multiplicativity at 3406 coprime pairs never imposed by the solver; $a_{1951} = \zeta_5^2$ to seven digits; and the completed L -function satisfies a functional equation of conductor 1951 with $\Gamma_{\mathbb{R}}(s+1)^2$ and root number $-\tau(\chi_{780})\zeta_5^3/\sqrt{1951}$. A scan in the spectral parameter confirms that $\lambda = 1/4$ is an eigenvalue rather than an assumption: the Hecke defect vanishes quadratically in R down to a floor that falls by a factor 2094 when the truncation is doubled, while the quadratic coefficient is stable to 0.5%. We emphasise at the outset that this is numerical evidence, not a proof; §5 states precisely what would be needed to close the gap.

1. INTRODUCTION

Artin’s holomorphy conjecture asserts that for every nontrivial irreducible complex representation ρ of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ the Artin L -function $L(\rho, s)$ is entire. In dimension 2 the conjecture is classified by the projective image, which is dihedral, A_4 , S_4 or A_5 , and by the parity $\det \rho(c) = \pm 1$, where c denotes complex conjugation.

All cases are settled except one. The dihedral case is classical (Hecke). The solvable cases A_4 and S_4 follow from Langlands–Tunnell, extending Maass’s construction of the relevant nonholomorphic forms. The odd icosahedral case is a theorem of Khare–Wintenberger together with the modularity lifting results of Buzzard–Dickinson–Shepherd–Barron–Taylor. What remains open is the *even icosahedral* case: $\det \rho(c) = +1$ with projective image A_5 .

The obstruction is structural. An even two-dimensional representation corresponds not to a holomorphic form but to a Maass form of Laplace eigenvalue exactly $1/4$, the bottom of the tempered spectrum. There is no known construction of such forms, and no deformation-theoretic route to them: every mod- λ representation attached to a holomorphic eigenform is odd, and parity is a twist invariant, so the modularity lifting machinery that resolves the odd case cannot be transported.

Doud and Moore [5] exhibited an even icosahedral representation of conductor 1951 and proved that 1951 is the least prime conductor of such a representation. Booker, Lee and Strömbergsson [2] list it, up to twist, among the even two-dimensional nondihedral Artin representations of conductor at most 2862, conditionally on Artin’s conjecture. Booker [1] developed an almost-monomiality criterion together with a variant of Turing’s method which

verifies holomorphy in a bounded region of the critical strip, under a working hypothesis on the zeros.

This paper takes the complementary route: rather than certifying $L(\rho, s)$ in a region, we attempt to *exhibit the automorphic object*. We report a numerical solution of the automorphy conditions for a Maass form of level 1951 and eigenvalue $1/4$ whose Hecke eigenvalues match ρ , and we subject it to a sequence of tests that the solver could not have anticipated.

Summary of the evidence. The solver receives only the level $N = 1951$, the character χ_{780} , the spectral parameter $R = 0$, a set of pullbacks for $\Gamma_0(1951)$, and a phase λ . It never sees the Doud–Moore quintic, the group A_5 , the golden ratio, or any Frobenius datum. What comes out:

- (1) 549/549 unramified primes below 4000: $|a_p/\chi(p)^3|$ agrees with the value forced by the factorisation type of the quintic modulo p (§4.1).
- (2) $a_n/\chi(n)^3$ is real and lies within 5.1×10^{-3} of $\mathbb{Z}[\varphi]$ at every one of the 3998 indices $n \leq 4000$ prime to 1951 (§4.2).
- (3) Hecke multiplicativity holds at 3406 coprime pairs, maximum error 8.8×10^{-5} ; only the single pair $(2, 3)$ entered the search (§4.3).
- (4) $a_{1951} = \zeta_5^2$ with $|a_{1951}| = 1.0000074$ (§4.4).
- (5) The completed L -function satisfies a functional equation of conductor 1951 with the *odd* Γ -factor; the even one fails decisively (§4.5).
- (6) The fitted phase is not a free parameter: it equals $\tau(\chi)\zeta_5^3/\sqrt{1951}$ to 7×10^{-4} degrees (§4.6).
- (7) $\lambda = 1/4$ is an eigenvalue, not an input (§4.7).

2. THE ARITHMETIC SIDE

2.1. The field. Let

$$f_5(x) = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344, \quad K = \mathbb{Q}[x]/(f_5).$$

Proposition 2.1. $\text{disc}(f_5) = 136178^2 \cdot 1951^4$, *exactly*. The polynomial f_5 has five real roots, so K is totally real of signature $(5, 0)$. Its Galois group is A_5 , and the index $[\mathcal{O}_K : \mathbb{Z}[x]] = 136178 = 2 \cdot 7 \cdot 71 \cdot 137$.

Proof. The discriminant is the resultant $\text{Res}(f_5, f'_5)$, an exact integer computation giving 268684727248076769842884, and $268684727248076769842884 = 136178^2 \cdot 1951^4$. The roots are -26.0876 , -3.4790 , -0.1047 , 1.2295 , 29.4417 , all real. Since the discriminant is a square, $\text{Gal} \subseteq A_5$; the factorisation types of f_5 modulo the 545 primes below 4000 at which f_5 is separable are exactly 1^5 , $(2, 2, 1)$, $(3, 1, 1)$ and 5 , with observed frequencies 0.0092, 0.2459, 0.3284, 0.4165 against the A_5 class densities $1/60$, $1/4$, $1/3$, $2/5$. The presence of both 3-cycles and 5-cycles forces $\text{Gal} = A_5$. $\square$

The four primes 2, 7, 71, 137 divide the index; at these f_5 is not separable modulo p and the splitting type must be read from `idealprimedec` rather than from the factorisation of f_5 . This is a standard trap and a silent one: a naive implementation assigns them the wrong classes.

Remark 2.2. Total reality is not incidental. For a two-dimensional ρ with $\det \rho(c) = +1$ one has $\rho(c)^2 = I$ and $\det \rho(c) = 1$, hence $\rho(c) = \pm I$; the projective image of c is therefore trivial and the A_5 -field is totally real. Proposition 2.1 thus confirms evenness independently of any automorphic computation.

2.2. The character and the twist. The group $(\mathbb{Z}/1951)^\times$ is cyclic of order $1950 = 2 \cdot 3 \cdot 5^2 \cdot 13$ and 3 is a primitive root. Let $\chi = \chi_{780}$ be the quintic character $\chi(n) = \zeta_5^{2 \log_3 n}$. It has order 5 and conductor 1951, and it is even, since $\chi(-1)^2 = 1$ and $\chi(-1)^5 = 1$ force $\chi(-1) = 1$. Consequently *any* form of nebentypus χ gives rise to an even representation.

Write $\rho = \rho_0 \otimes \chi^3$. Then $\det \rho_0 = \det \rho \cdot \chi^{-6} = 1$, so ρ_0 takes values in $SL_2(\mathbb{C})$ with image the binary icosahedral group $2.A_5 \cong SL(2, 5)$. Its traces lie in

$$T = \{0, \pm 1, \pm \varphi, \pm \varphi^{-1}, \pm 2\}, \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad (1)$$

and in particular $a_n = \chi(n)^3 t_n$ with t_n real and in $\mathbb{Z}[\varphi]$.

2.3. The local datum at 1951. The ramification is tame, since $5 \mid 1950$. That the conductor is 1951 rather than 1951^2 says the conductor exponent is 1, i.e. $\dim V^I = 1$: inertia acts trivially on a line and by a quintic character on a complement. Hence a_{1951} is the Frobenius eigenvalue on V^I , a root of unity. This is a sharp, falsifiable prediction, tested in §4.4.

Remark 2.3. The two central lifts ρ_0 and $\rho_0 \otimes \eta$, η the quadratic character modulo 1951, differ in parity, since $1951 \equiv 3 \pmod{4}$ makes η odd. They also differ in conductor: twisting by η destroys the inertia-fixed line and raises the conductor to 1951^2 . Exactly one lift has conductor 1951, and the computation of §4.5 identifies it as the one with $\rho(c) = -I$.

3. THE COMPUTATION

We solve for a Maass form

$$f(z) = \sqrt{y} \sum_{n \neq 0} c_n K_{iR}(2\pi|n|y) e(nx), \quad f(\gamma z) = \chi(d)f(z) \text{ for } \gamma \in \Gamma_0(1951),$$

at $R = 0$, by Hejhal's algorithm: sample a horocycle at height Y , pull back into the fundamental domain, impose automorphy, and solve the resulting linear system for $c_{\pm 1}, \dots, c_{\pm M}$.

Three points of the implementation deserve comment.

3.1. The cusp-zero relation. At level 1951 the cusp 0 has width N , and a census of 2000 sample points found 1998 of them pulling back to a neighbourhood of 0. The transformation there is therefore not a detail but the dominant constraint. For a complex nebentypus the Fricke involution does *not* act by a sign: $f|W_N$ lies in $L^2(\Gamma_0(N), \bar{\chi})$, and

$$f(\sigma_0 z) = \lambda_N f^*(z), \quad a_n(f^*) = \overline{a_n(f)}.$$

The relation is antilinear. Writing $\hat{c} = u + iv$ turns the $2M$ complex unknowns into a real $4M \times 4M$ system

$$\begin{pmatrix} I - \operatorname{Re} S & \operatorname{Im} D \\ -\operatorname{Im} S & I - \operatorname{Re} D \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = 0, \quad S = A + \lambda B, \quad D = A - \lambda B,$$

normalised by $\hat{c}_1 = d_1$, $\operatorname{Im} \hat{c}_1 = 0$. Using ± 1 in place of the antilinear relation — the correct rule for trivial character — is wrong at 1998 of 2000 points.

3.2. Truncation and precision. Per-point truncation $M_m = \lceil 43/(2\pi y^*) \rceil$ reduces the cost by a factor of 64 at this level and makes $N = 1951$ feasible at all. The double precision limit is governed by d_1/d_M ; at $M = 8000$, $Y = 4.35 \times 10^{-4}$ this ratio is 3.2×10^9 , comfortably inside the usable range, while $M = 14673$ would put it at 8×10^{18} and destroy the solution.

3.3. The K -Bessel function. Standard compiled routines for K_{iR} fail at small R with small argument. We use instead

$$K_{iR}(x) = \int_0^\infty e^{-x \cosh t} \cos(Rt) dt$$

evaluated by the trapezoidal rule. The integrand is analytic and bounded in the strip $|\operatorname{Im} t| < \pi/2$, so the rule converges like $e^{-\pi^2/h}$; at $h = 0.05$ this is past double precision with about 300 nodes. Tabulating $K_{iR}(x)e^x$, which is $O(1)$ and slowly varying, on a uniform grid in $\log x$ and interpolating by four-point Lagrange cubics gives 4×10^{-15} at 1.2×10^7 evaluations per second. Against `mpmath` the maximum relative error over the parameter range used here is 6.5×10^{-16} .

4. VERIFICATION

Throughout, $t_n := a_n/\chi(n)^3$.

4.1. Frobenius classes. For each prime $p < 4000$ with $p \nmid 136178 \cdot 1951$ the factorisation type of f_5 modulo p determines the projective conjugacy class of Frob_p , hence $|t_p|$:

$$1^5 \mapsto 2, \quad (2, 2, 1) \mapsto 0, \quad (3, 1, 1) \mapsto 1, \quad 5 \mapsto \varphi \text{ or } \varphi^{-1}.$$

The four index divisors are supplied by `idealprimedec`.

Theorem 4.1. *At all 549 primes $p < 4000$ with $p \neq 1951$, the computed $|t_p|$ agrees with the value forced by the factorisation type of f_5 modulo p , with maximum error 3.6×10^{-3} and root-mean-square error 5.7×10^{-4} . Restricted to $p \leq 2000$, where the coefficients are better resolved, the maximum error is 5.1×10^{-5} over 302 primes.*

The 5-cycles split 112 to 115 between $|t_p| = \varphi$ and $|t_p| = \varphi^{-1}$, against the 1 : 1 ratio forced by A_5 , and the split agrees with an independent `idealprimedec` determination at all 34 inert primes below 500. The five split primes below 4000 are 563, 2137, 2711, 2741, 2791, at which $t_p = +2.00000$: the Ramanujan bound is saturated exactly where the group theory says it must be.

4.2. Untwisting.

Theorem 4.2. *For all 3998 indices $n \leq 4000$ prime to 1951,*

$$|\operatorname{Im} t_n| \leq 4.2 \times 10^{-3}, \quad \operatorname{dist}(\operatorname{Re} t_n, \mathbb{Z}[\varphi]) \leq 5.1 \times 10^{-3},$$

with root-mean-square values 4.1×10^{-4} and 4.5×10^{-4} .

Nothing in the solver refers to χ^3 , to A_5 , or to φ . The dyadic maxima of $|a_n|$ are $1, \varphi, \varphi^2, \varphi^3$ exactly, out to $n = 4096$; past that the coefficients enter the truncation noise of the expansion.

4.3. Hecke relations.

Theorem 4.3. *Over the 3406 coprime pairs (m, n) with $mn \leq 2000$, $|a_m a_n - a_{mn}| \leq 8.8 \times 10^{-5}$. The recursions $a_{p^2} = a_p^2 - \chi(p)$ and $a_{p^3} = a_p a_{p^2} - \chi(p) a_p$ hold with maximum errors 3.4×10^{-5} and 6.2×10^{-5} .*

Only the single relation $|a_2 a_3 - a_6|$ was ever used, as the objective of a one-dimensional search.

4.4. The ramified prime. The computed value is

$$a_{1951} = -0.809027460 + 0.587783429i, \quad |a_{1951}| = 1.0000074, \quad a_{1951}^5 = 1.0000370 + 0.0000381i,$$

that is, $a_{1951} = \zeta_5^2$ to seven digits. This is exactly the prediction of the local analysis: conductor exponent 1 forces $|a_{1951}| = 1$, and the tame quintic ramification forces a fifth root of unity.

4.5. Parity and the functional equation. The reflection $T : f(z) \mapsto f(-\bar{z})$ preserves $L^2(\Gamma_0(N), \chi)$: for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ one has $-\bar{\gamma}z = \gamma'(-\bar{z})$ with $\gamma' = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix} \in \Gamma_0(N)$ of the same lower-right entry. Hence parity is well defined, and the solver, which never imposes it, returned

$$\max_{n \leq 2000} |c_{-n} + c_n| = 6.0 \times 10^{-5}, \quad \max_{n \leq 2000} |c_{-n} - c_n| = 8.5.$$

The form is *odd*. By the correspondence this says $\rho(c) = -I$ and Γ -factor $\Gamma_{\mathbb{R}}(s+1)^2$. It remains an even Artin representation: $\det \rho(c) = \chi(-1) = +1$, in agreement with the total reality of K .

Set $\Lambda(s) = N^{s/2} \Gamma_{\mathbb{R}}(s+\epsilon)^2 L(s)$ and $A_{\epsilon}(y) = \sum_{n \geq 1} a_n \phi_{\epsilon}(ny/\sqrt{N})$, where ϕ_{ϵ} is the inverse Mellin transform of $\Gamma_{\mathbb{R}}(s+\epsilon)^2$, namely $\phi_0(t) = 4K_0(2\pi t)$ and $\phi_1(t) = 4tK_0(2\pi t)$. The functional equation $\Lambda(s) = w\bar{\Lambda}(1-s)$ is equivalent to $A_{\epsilon}(y) = w y^{-1} B_{\epsilon}(1/y)$ with B built from $\bar{a}_n$.

Theorem 4.4. *With $\epsilon = 1$ and conductor 1951, using only the coefficients a_n with $n \geq 1$,*

$$|w(y)| = 1 \pm 2 \times 10^{-6}, \quad \arg w(y) = -116.7522^\circ,$$

constant to 1.8×10^{-6} for $y \in [0.55, 1.9]$. With $\epsilon = 0$ the same computation gives $|w(y)|$ ranging over $[0.859, 1.143]$.

The even Γ -factor is excluded; the odd one holds to the precision of the coefficients.

4.6. The phase was never free. The phase $\theta = \arg \lambda$ was located by golden-section minimisation of $|a_2 a_3 - a_6|$ over an interval of 35° , a criterion with no knowledge of Gauss sums. It converged to $\theta = 63.24805^\circ$. Independently, Theorem 4.4 gives a root number of argument -116.7522° , so that $w = -e^{i\theta}$, the sign being the archimedean factor $i^{1+1} = -1$ of the odd Γ -factor. Finally

$$\frac{\tau(\chi)}{\sqrt{1951}} = -0.889038556925 - 0.457832332084i, \quad \arg = -152.752680^\circ,$$

and $-152.752680 + 216 = 63.247320$. That is,

$$e^{i\theta} = \frac{\tau(\chi)}{\sqrt{1951}} \cdot \zeta_5^3 \quad \text{to } 7.3 \times 10^{-4} \text{ degrees.} \quad (2)$$

The admissible phases form a set of five, the ambiguity being precisely the additive-character convention in $\varepsilon(\chi, \psi) = \tau(\chi)/\sqrt{p}$. A search over a continuum landed on one of them, and the same value reappears as the root number of an L -function the search never saw.

4.7. $\lambda = 1/4$ is an eigenvalue. Everything above is conditional on $R = 0$, which was an input. Hejhal's method certifies an eigenvalue not by the residual — which is $\approx 3 \times 10^{-15}$ at every R and detects nothing — but by the collapse of relations the solver does not impose. We therefore scan R with λ pinned to the exact value (2), so that R is the only free quantity, and measure

$$C_1 = |a_2 a_3 - a_6|, \quad C_2 = |a_2 a_5 - a_{10}|, \quad C_3 = |a_2^2 - a_4 - \chi(2)|, \quad D = \left(\frac{1}{15} \sum_{p < 50} \text{dist}(t_p, T)^2 \right)^{1/2}.$$

R	C_1 ($M=4000$)	C_1/R^2	C_1 ($M=8000$)	C_1/R^2
0	6.8257×10^{-8}	—	3.2589×10^{-11}	—
0.0005	1.6495×10^{-5}	65.98	1.6379×10^{-5}	65.52
0.0015	1.4812×10^{-4}	65.83	1.4745×10^{-4}	65.53
0.0040	1.0547×10^{-3}	65.92	1.0503×10^{-3}	65.64
0.0090	5.3806×10^{-3}	66.43	5.3587×10^{-3}	66.16
0.0180	2.2156×10^{-2}	68.38	2.2078×10^{-2}	68.14
0.0320	7.6342×10^{-2}	74.55	7.6193×10^{-2}	74.41
0.0550	2.9619×10^{-1}	97.91	2.9719×10^{-1}	98.25
0.0820	1.2523	186.2	1.2753	189.7

TABLE 1. The Hecke defect against the spectral parameter, at two truncations. C_1/R^2 is flat across an eighteenfold range in R and a threehundredfold range in C_1 . The log-log slopes between consecutive small- R points are 1.998, 2.001, 2.009, 2.042.

Only $R \geq 0$ need be scanned: K_{iR} depends on R^2 , so $C(R) = C(-R)$ and $R = 0$ is stationary whatever the truth. A dip at 0 is therefore not by itself evidence. What discriminates is the behaviour as $R \rightarrow 0$: since C is even and non-negative, $C(R) = C_0 + aR^2 + O(R^4)$, and a non-eigenvalue has $C_0 > 0$, so that C plateaus. An eigenvalue has $C_0 = 0$ and C falls as aR^2 all the way to the numerical floor.

Theorem 4.5. *The data of Table 1 are consistent with $C_0 = 0$ and inconsistent with any C_0 above 4.2×10^{-8} at $M = 4000$. Doubling the truncation to $M = 8000$ reduces $C_1(0)$ from 6.83×10^{-8} to 3.26×10^{-11} , a factor of 2094, while the quadratic coefficient moves from 66.51 to 66.20, a shift of 0.47%. All four detectors sit at $\approx 3 \times 10^{-11}$ at $R = 0$ and $M = 8000$.*

Justification. If $C_0 > 0$ then $C_1/R^2 = C_0/R^2 + a + O(R^2)$ must diverge as $R \rightarrow 0$; at $R = 5 \times 10^{-4}$ the term C_0/R^2 equals $4.0 \times 10^6 C_0$. The observed change in C_1/R^2 between the two smallest R is 0.149, whence $C_0 \leq 4.2 \times 10^{-8}$, the same order as the measured value at $R = 0$. Extrapolating $C_0 + aR^2 + bR^4$ from $R \geq 5 \times 10^{-4}$ alone, never using the $R = 0$ row, predicts the $R = 0$ values: 4.62×10^{-8} against a measured 6.83×10^{-8} for C_1 , and 1.948×10^{-6} against 1.955×10^{-6} for C_3 , an agreement of 0.4%. The residue at $R = 0$ is thus a smooth, R -independent systematic — the discretisation error of the scheme — and the M -dependence confirms it: it falls with M while the spectral coefficient does not. $\square$

5. WHAT THIS IS, AND WHAT IT IS NOT

Scope of the claim. The computation reported here is carried out in double-precision floating point. It is not a proof of Artin’s holomorphy conjecture for the conductor-1951 representation, and nothing above should be read as one.

To convert the evidence into a theorem, two things are required, in this order.

First, rigorous existence. One must show that the numerical solution lies within a certified distance of a genuine Maass form. The method of Booker, Strömbergsson and Venkatesh supplies *a posteriori* bounds of exactly this kind: given an approximate eigenfunction with a small residual in a suitable norm, it certifies the existence of a true eigenvalue nearby. That argument has to be run in interval arithmetic, with explicit control of the pullback, of the

truncation, and of the K -Bessel evaluation. The quadrature of §3.3 is well suited to this: its error is bounded analytically by $e^{-\pi^2/h}$ rather than empirically.

Second, identification. A certified Maass form with eigenvalues close to $\text{tr } \rho(\text{Frob}_p)$ is not yet the same object as ρ . Closing that step needs a rigidity statement — a converse theorem, or strong multiplicity one applied to a finite set of primes with an effective bound — turning agreement at finitely many primes into equality of L -functions. This is the substantive mathematical gap, and it is where the interesting work lies.

Independently of either, the algebraic content of §2 is exact and is formally verifiable: the discriminant identity, the total reality, the Galois group, the order and parity of χ_{780} , the Frobenius cycle types at every prime below 4000, and the trace set (1). A companion Lean 4 development formalises these, and isolates the analytic input above as a single named hypothesis, so that precisely what is assumed can be read off from the statement rather than inferred from prose.

Remark 5.1. One caveat deserves recording. Booker, Lee and Strömbergsson list two representations of conductor 1951, both marked as not twist-minimal, and their framework carries both parities through the symbol $\chi^{(\epsilon)}$, $\epsilon \in \{0, 1\}$. We were unable to determine from the published tables which ϵ the conductor-1951 entries occupy. Our computation asserts $\epsilon = 1$; §4.5 is the evidence, and a reader with access to the underlying data should check it.

DATA AND CODE

The solver, the verification scripts, the K -Bessel validation against mpmath, and the coefficient files at $M = 4000$ and $M = 8000$ accompany this note.

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